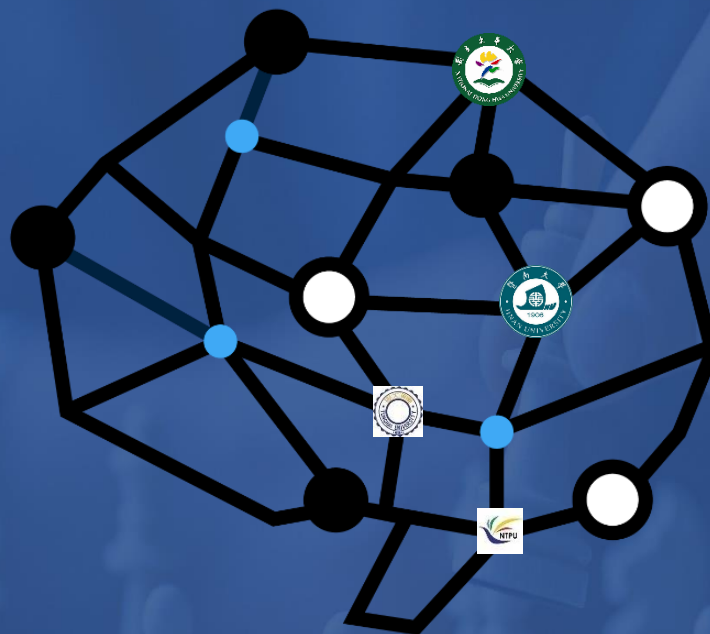




Chap. 6 Monte Carlo Tree Search

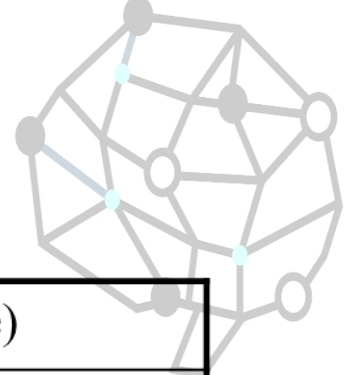
課程：人工智慧導論

授課教師：周信宏

AlphaGo

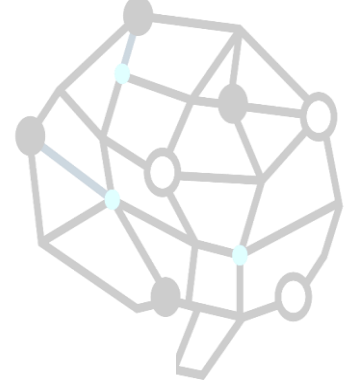


Complexity of Game



Games	Log(state-space)	Log(game tree size)
Nine Men's Morris	10	12
Connect-four	14	21
Checkers	21	31
Othello	28	58
Chess	46	123
Chinese chess	48	150
Hex(11x11)	57	98
Shogi	71	226
Renju(15x15)	105	70
Go-Moku(15x15)	105	70
Go(19x19)	172	360

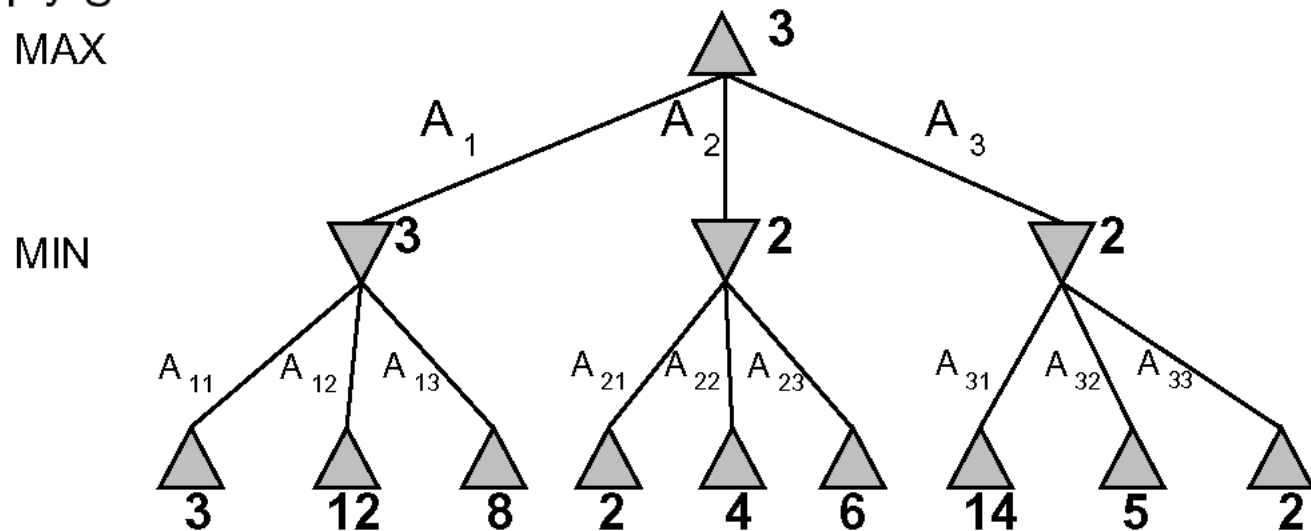
Game tree



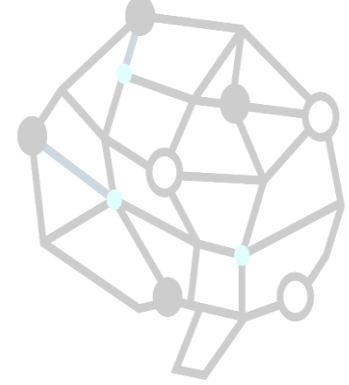
Perfect play for deterministic, perfect-information games

Idea: choose move to position with highest *minimax value*
= best achievable payoff against best play

E.g., 2-ply game:



Evaluation function of chess



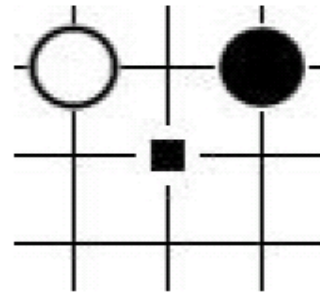
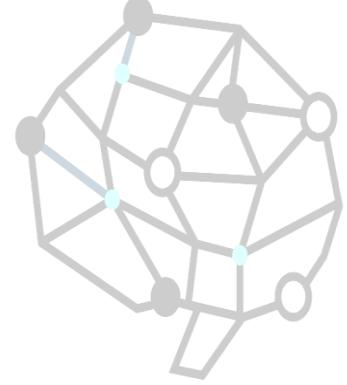
- Evaluation function



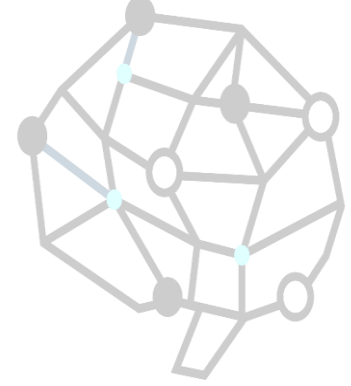
Piece	Value
Pawn	100
Knight	320
Bishop	325
Rook	500
Queen	975
King	32767

Traditional methods for Go Program

- Human-like Go Program
 - Pattern
 - Rule-based

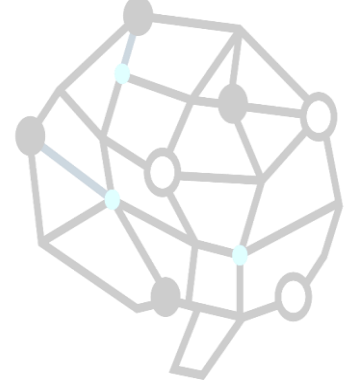


Introduction to MCTS

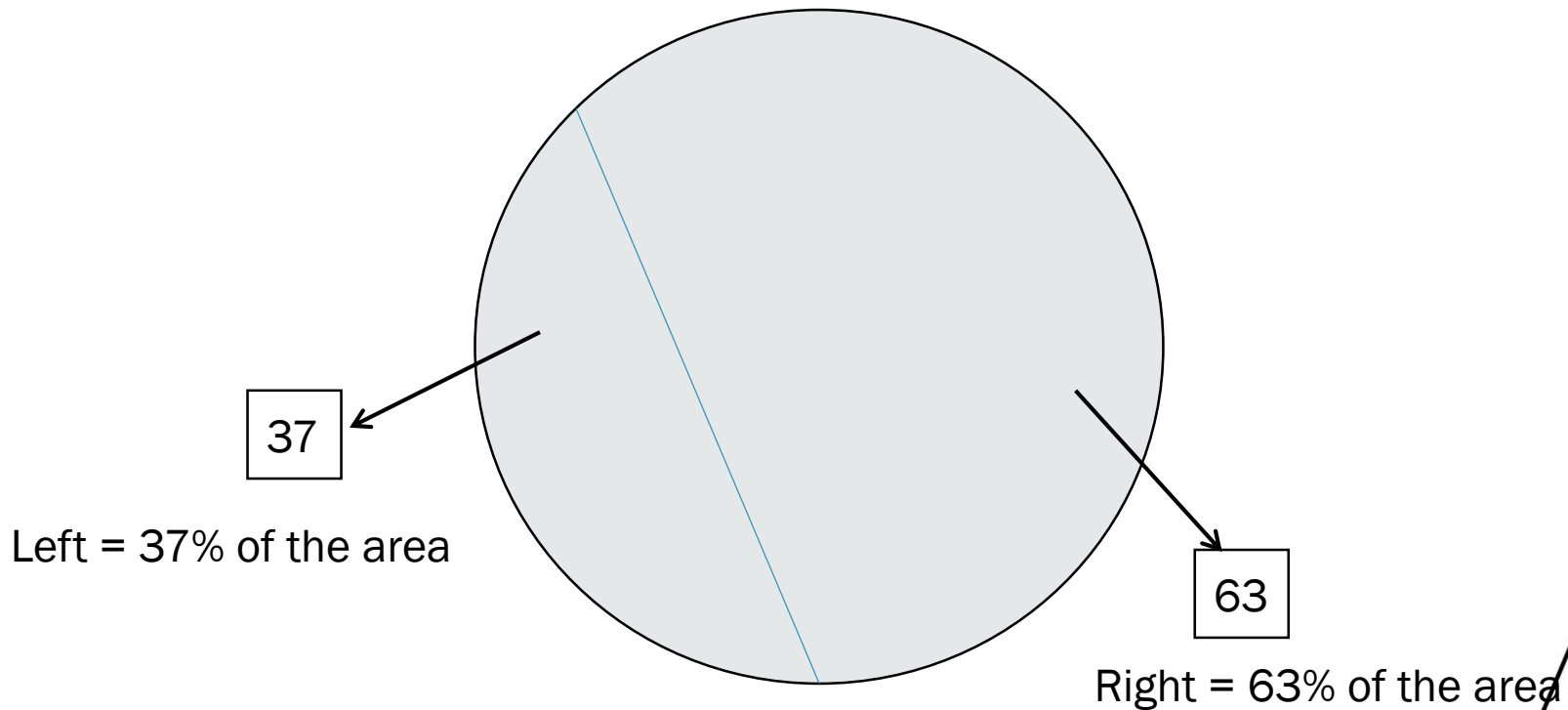


- A best-first search algorithm
 - Greedy search, A* search
- Using **Monte Carlo Method** to evaluate moves
- Building an unbalancing tree in memory
- Anytime algorithm

Monte Carlo Method

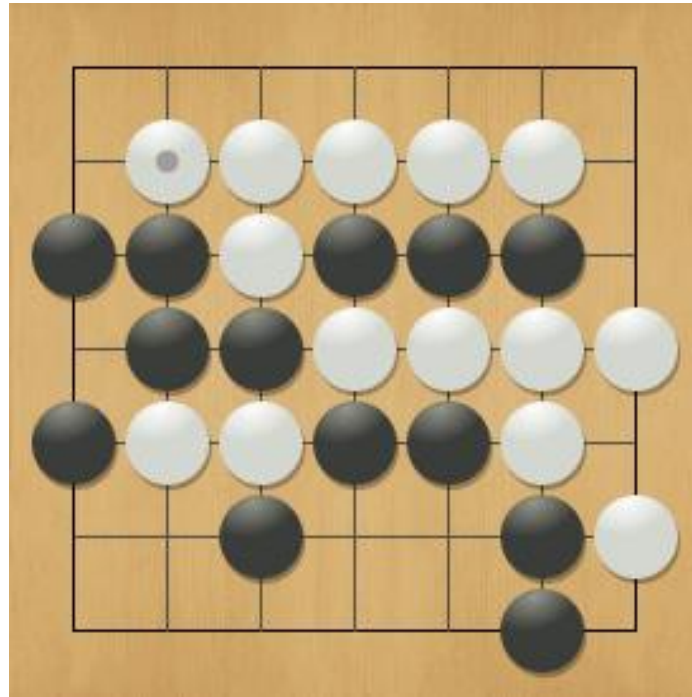


If we randomly place one hundred of beans...

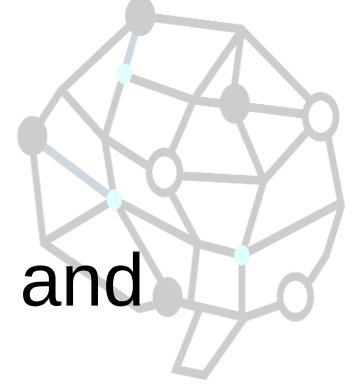


Monte Carlo Method

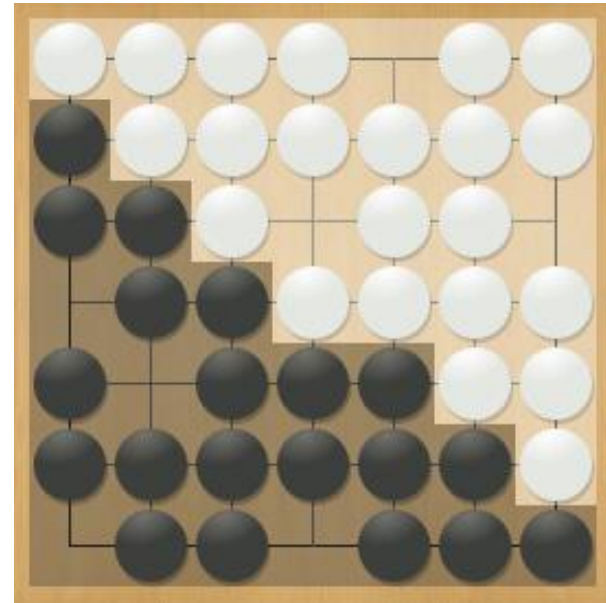
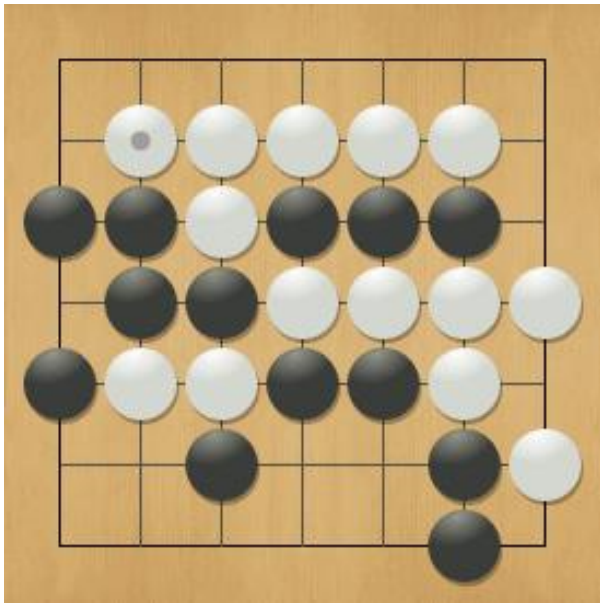
- Simulate one time = play a game, and get the result



Monte Carlo Method



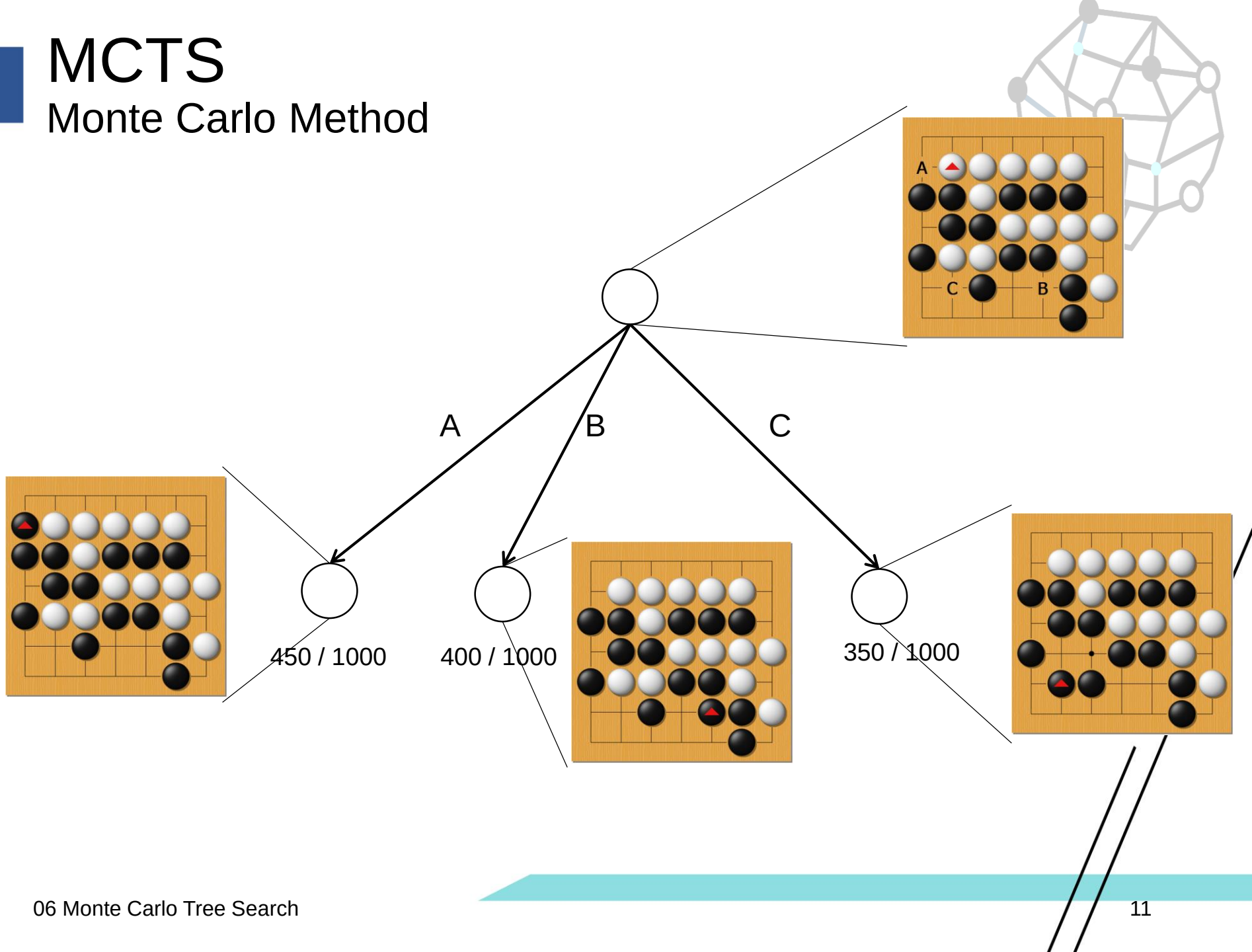
- Measure the situation : simulate N times and calculate the winning rate



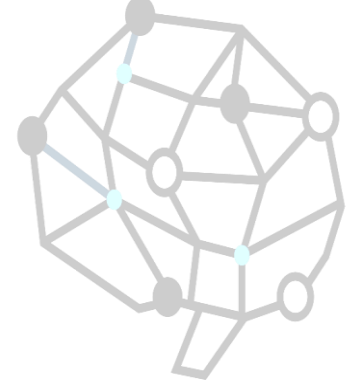
Winning rate = $45 / 100 = 0.45$

MCTS

Monte Carlo Method

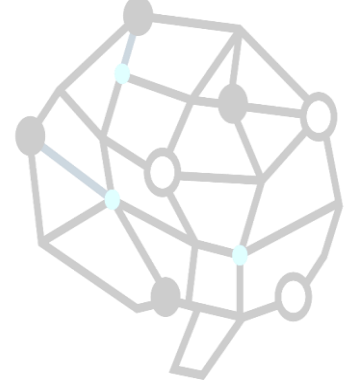


Upper Confidence Bound



- Exploitation(開發): Select next point by previous behavior
- Exploration (探險): Try other point
- UCB : to consider the balance between exploration and exploitation

Upper Confidence Bound



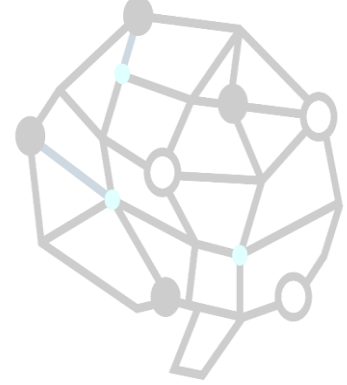
$$UCB_i = X_i + \sqrt{\frac{2 \ln N}{ExEValue * T_i}}$$

- X_i : the winning rate of the node i,

winning rate = wins / visits

- N : visits of parent of node i.
- T_i : visits of node i.

Upper Confidence Bound

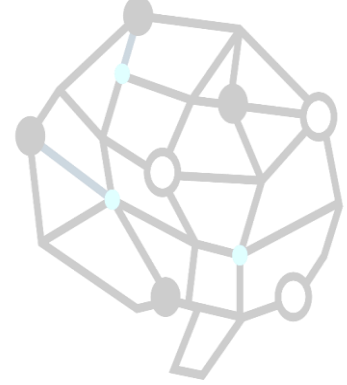


$$UCB_i = X_i + \sqrt{\frac{2 \ln N}{ExEValue * T_i}}$$

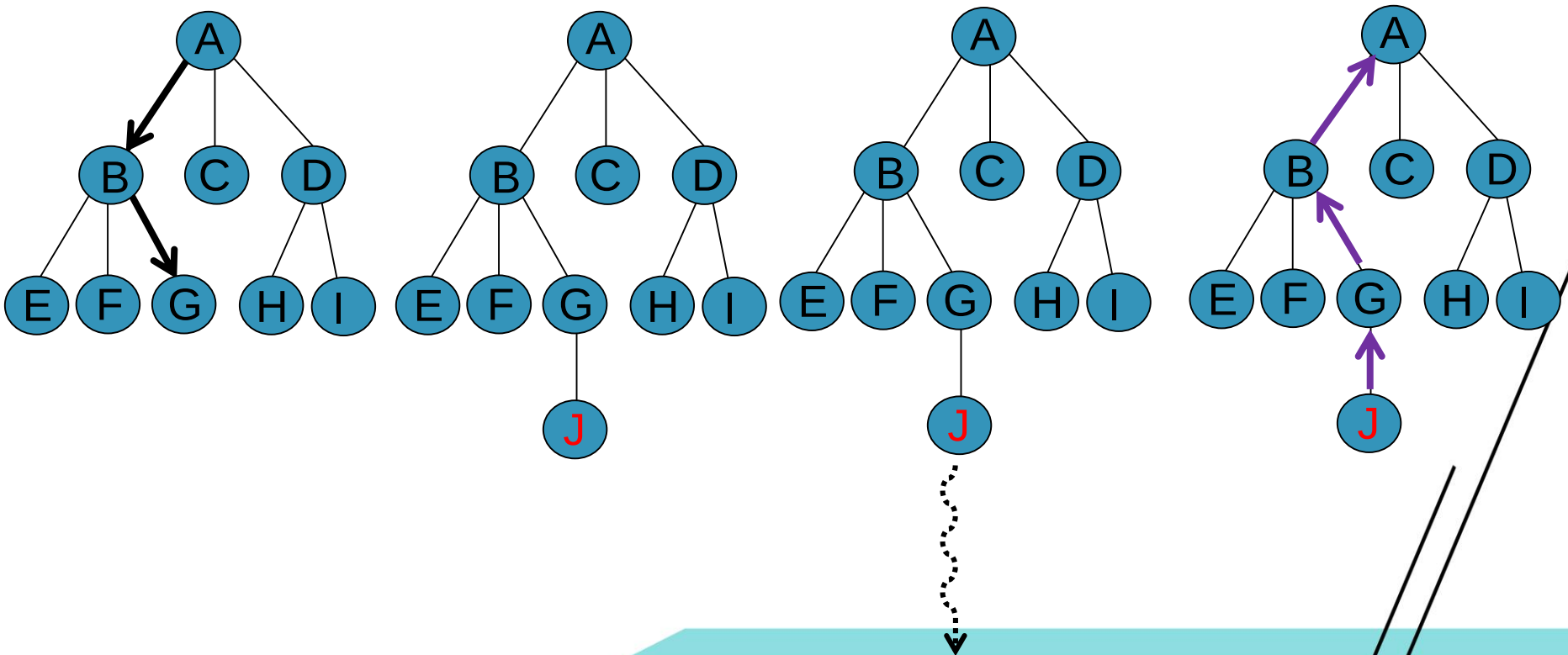
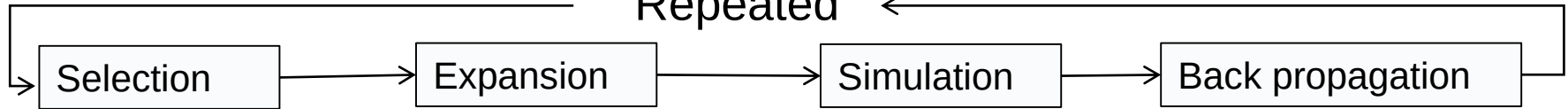
	A	B	C
visits	27	20	50
wins	6	5	15
X_i	0.22	0.25	0.3
Tuning Value	0.18	0.21	0.13
UCB Value	0.40	0.46	0.43



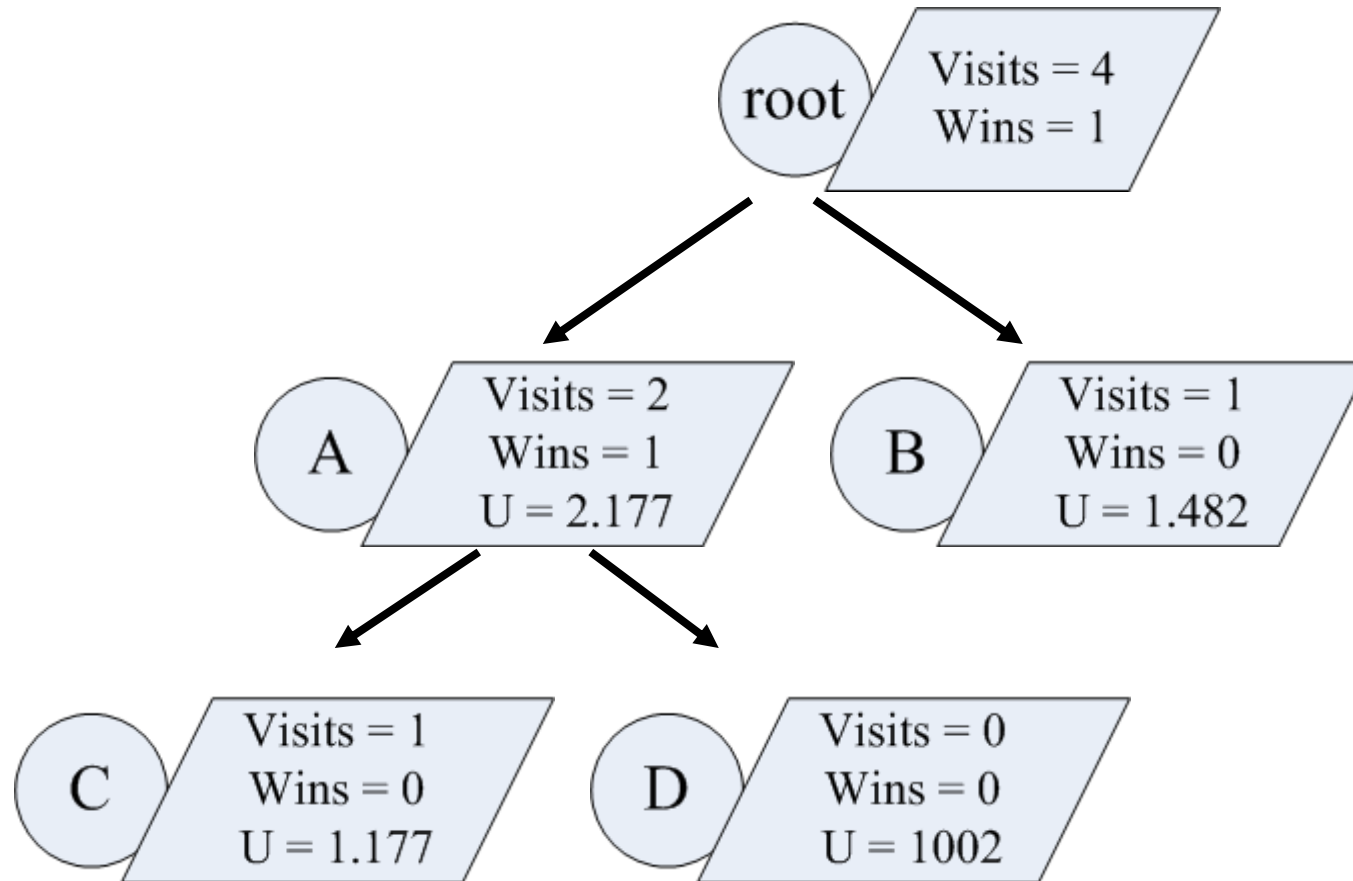
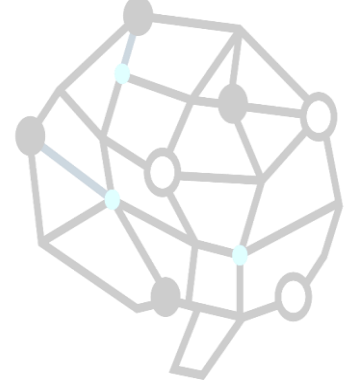
Monte-Carlo Tree Search(2007, Herik)

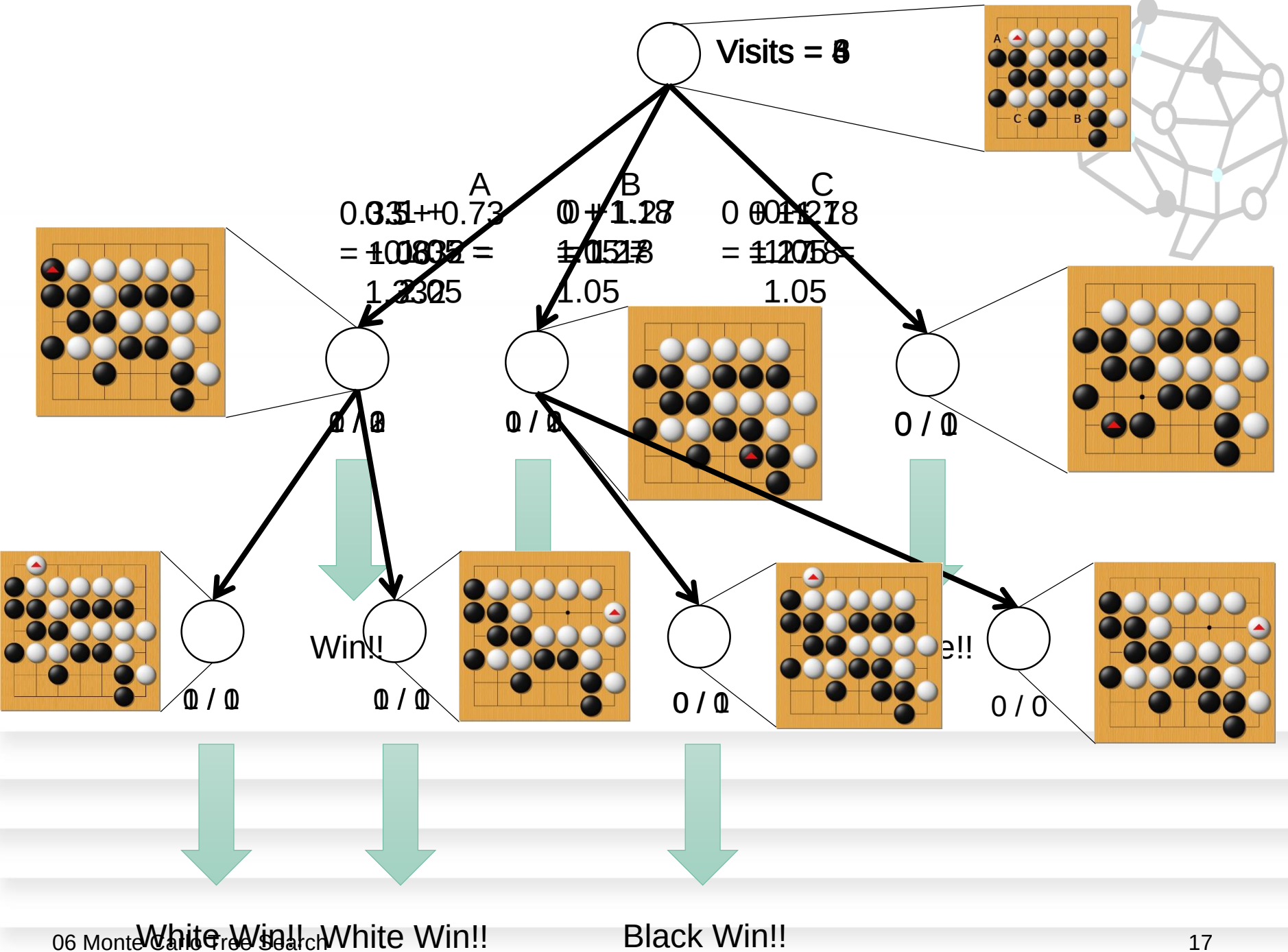


Repeated

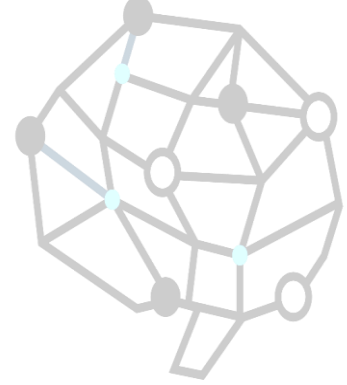
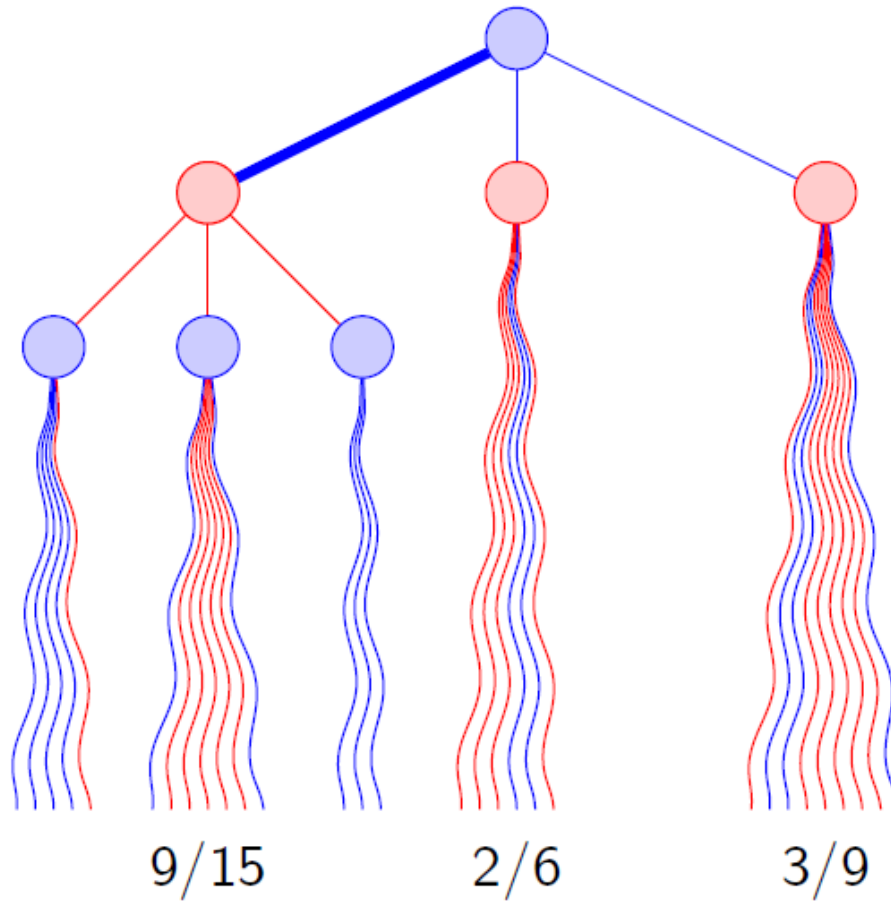


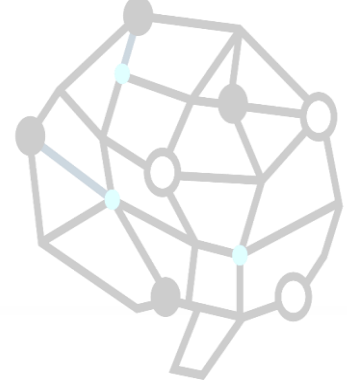
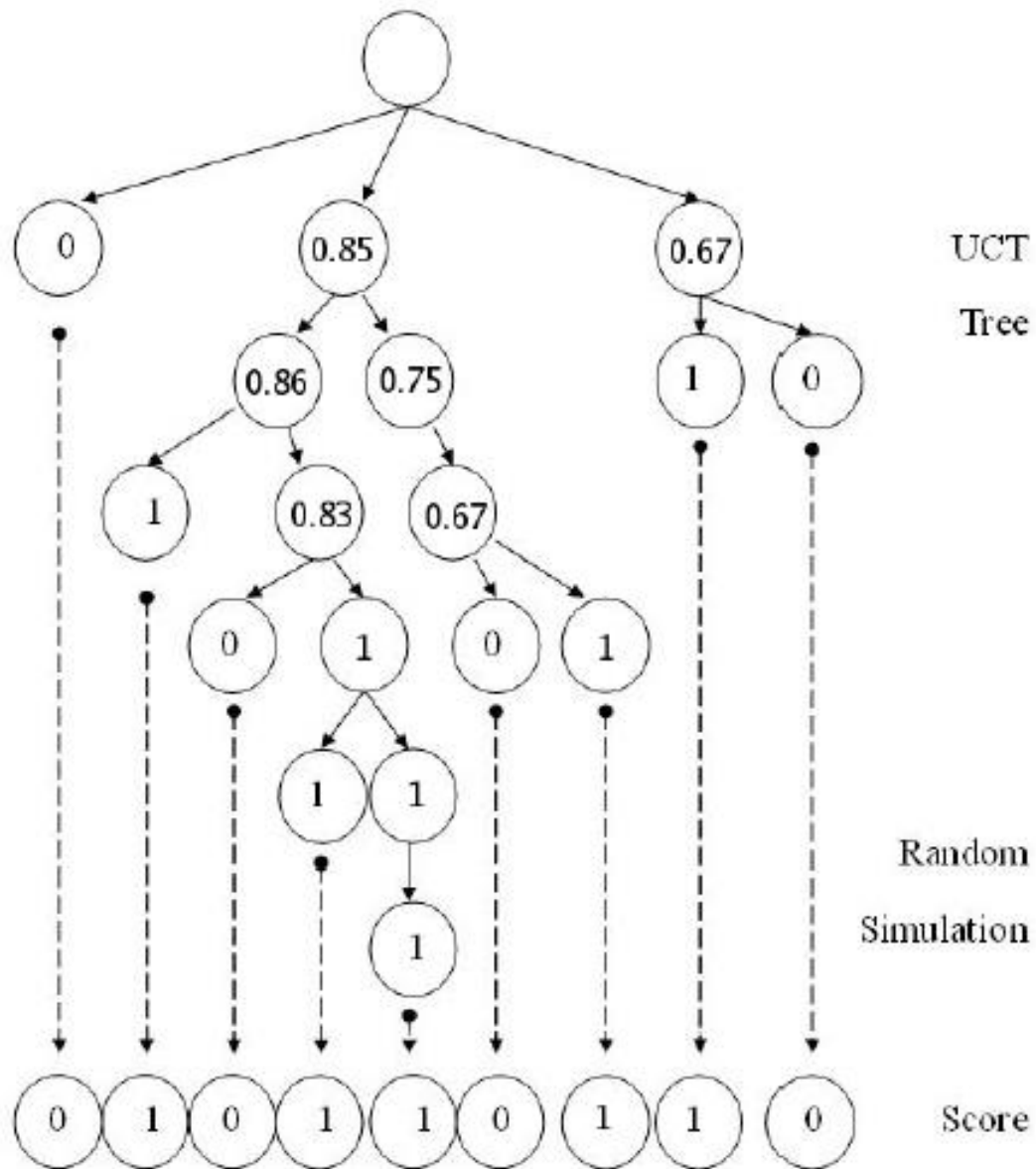
Upper Confidence Bound for Tree Search





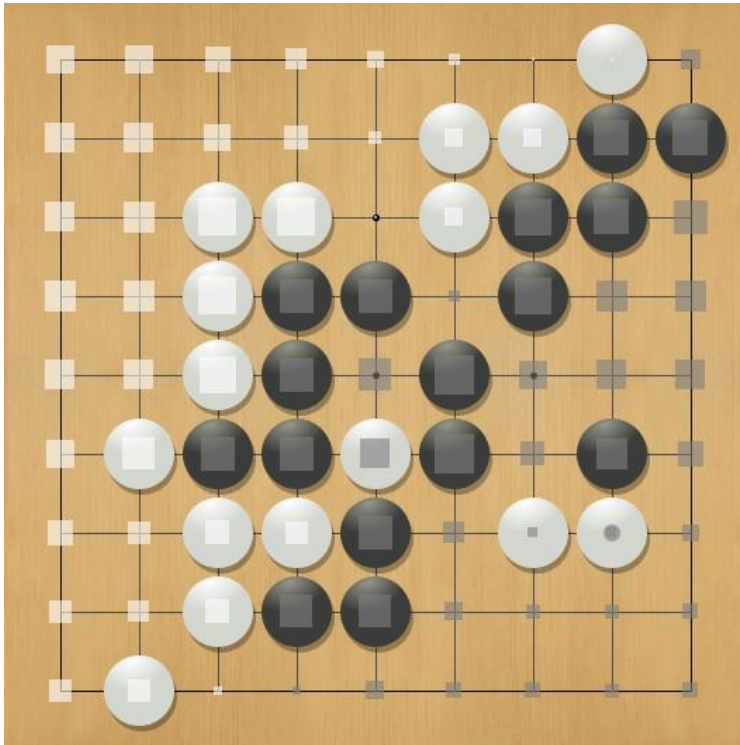
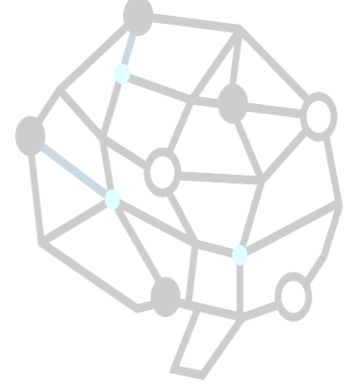
UCT-不平衡的樹



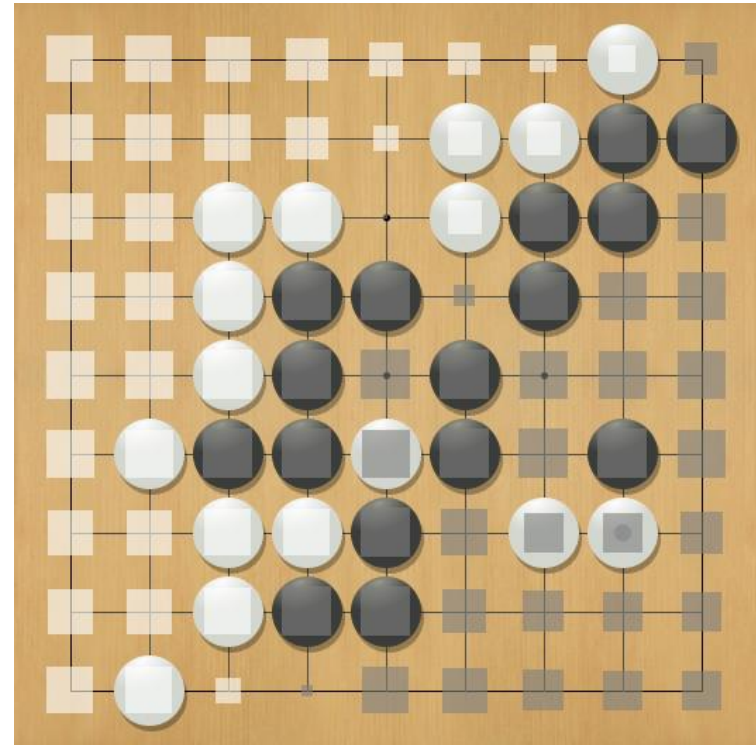


Simulation

- Goal : measure the situation correctly



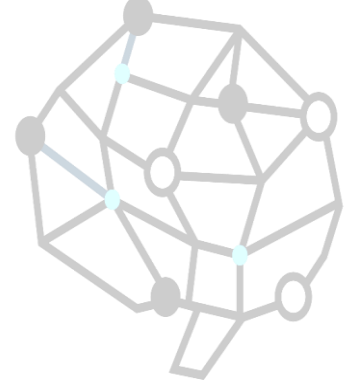
All random



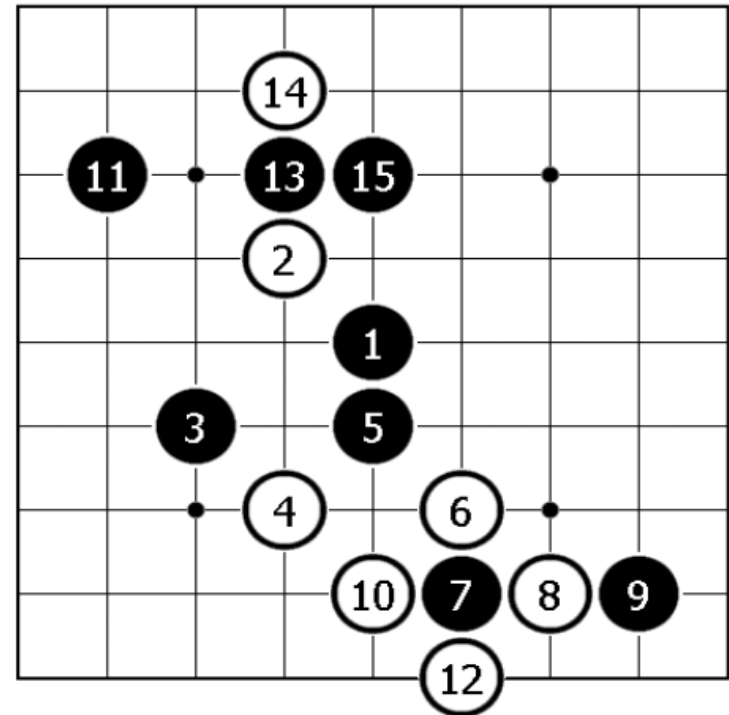
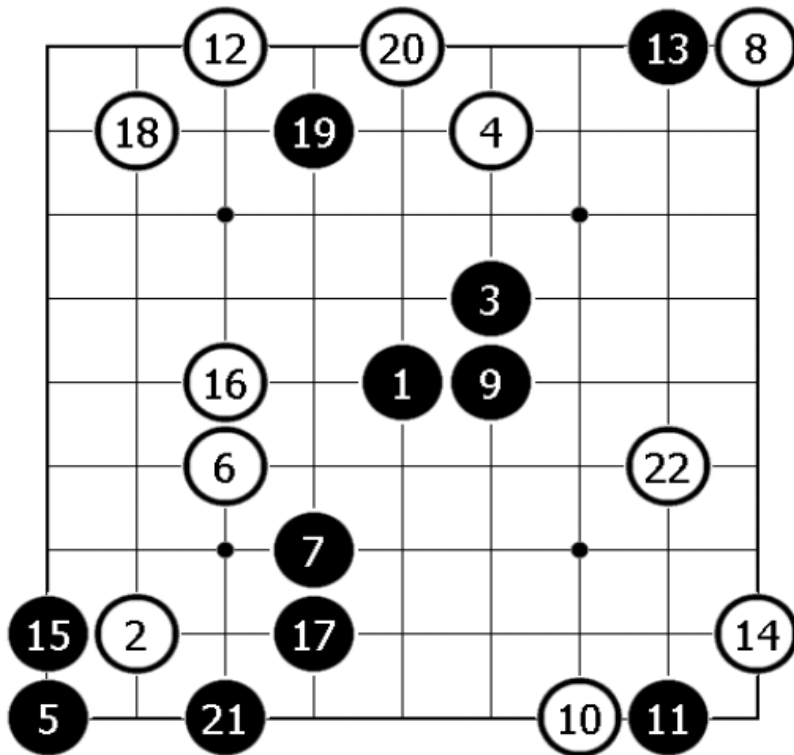
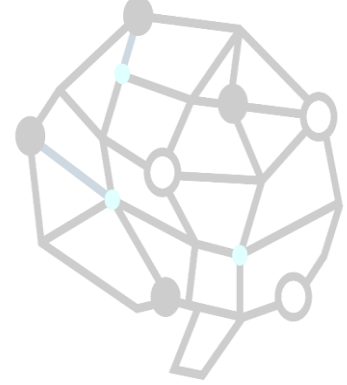
Using Heuristic

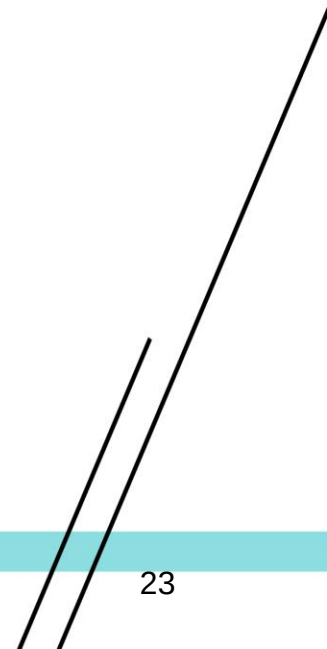
Simulation

- Probability Distribution
- Random
- Rule-based
 - 1. Capture
 - 2. Escape
 - 3. Pattern
 - 4. Random

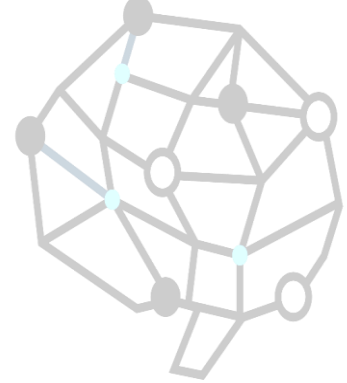
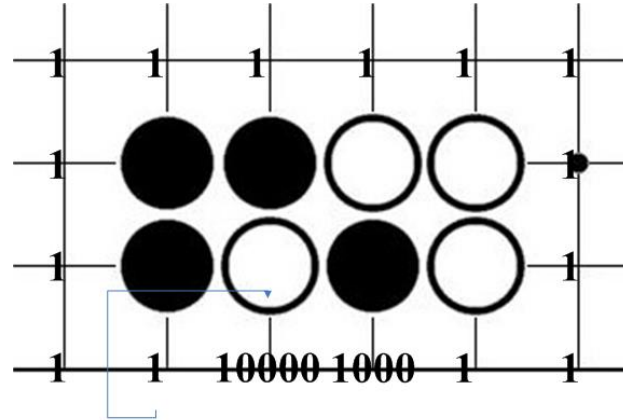
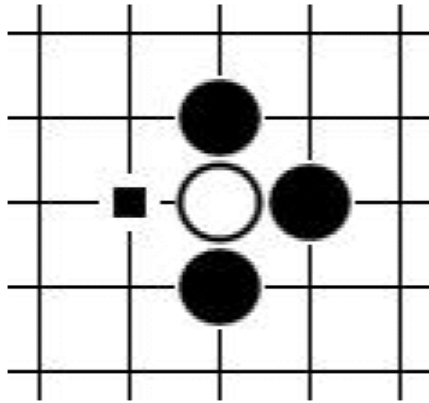


Random v.s. rule base

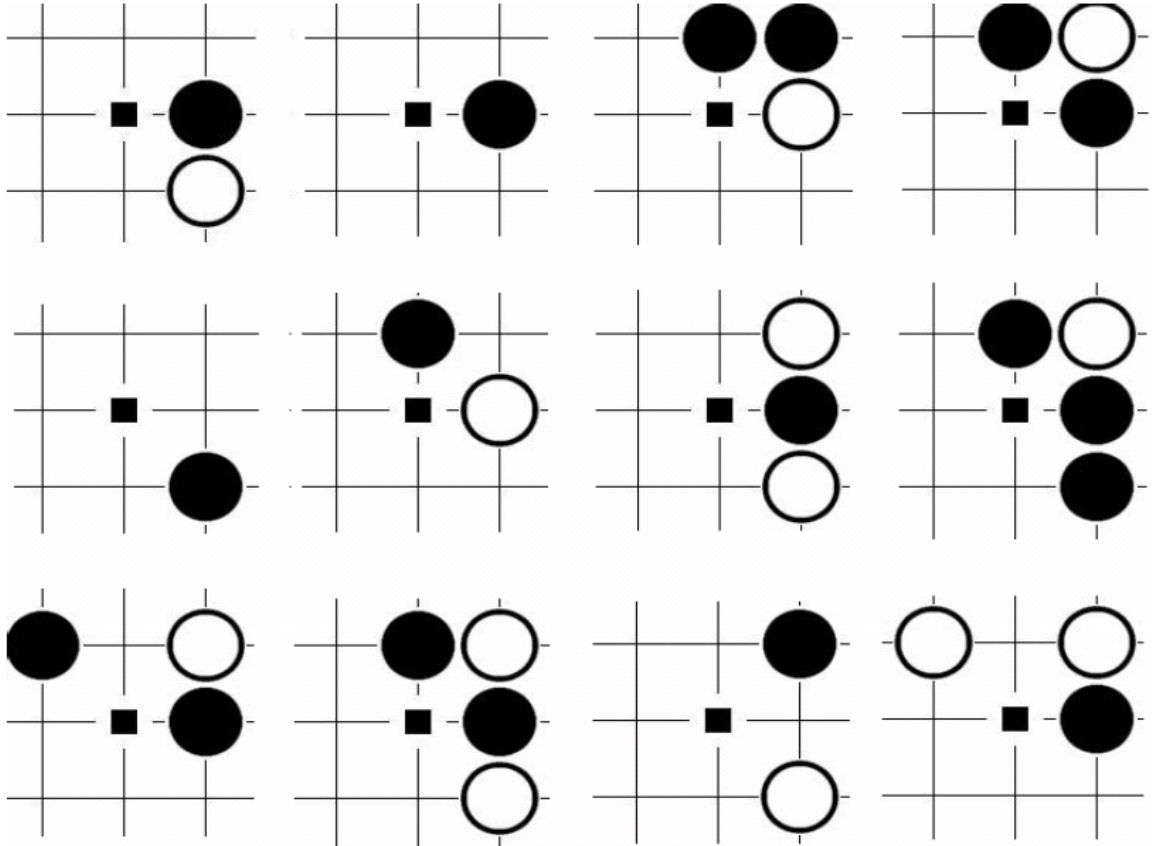
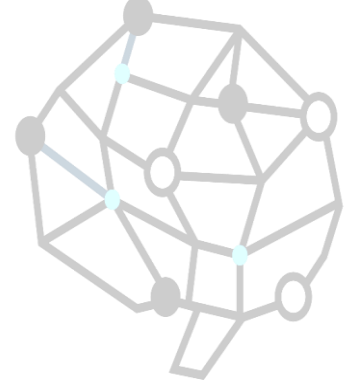




Rule based



Pattern



實驗結果



Rule	Pattern	Winrate
X	X	32%
O	X	44%
O	O	63%

A hand is shown moving a white chess piece on a chessboard. The entire image is overlaid with a semi-transparent blue filter. The text "THE END." is centered in the middle of the image, with a white horizontal line underneath it.

THE END.
